

Are We Alone? The Search for Earth 2.0

Authors: Tanishq Dahiya, Kartik Shahrawat

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Abstract

This assignment goes deeper into one of the most profound questions in astronomy: "Are we alone?" By analyzing the physical characteristics of thousands of exoplanets, we aim to determine how common "Earth-like" worlds are compared to the vast population of uninhabitable gas giants. We used the official NASA Exoplanet Archive dataset [1] to visualize the relationship between planetary radius and orbital distance. Our analysis reveals that while the majority of discovered planets are hostile environments, a distinct, 1 potentially habitable planet exists. We visualized this phenomenon using a scatter plot which was further log scaled for better visualization. With a specialized inset zoom-in feature to label the top unique candidates in the "Goldilocks Zone."

1. Datasets

We acquired the dataset for this study from the NASA Exoplanet Archive, a comprehensive public database maintained by the California Institute of Technology. The initial raw dataset contained over 35,000 rows of data. Each row contains high-dimensional data with over 90 columns of astrophysical parameters

2. Data Exploration, Processing, and Cleaning

To prepare the dataset for visualization, we used Python with the Pandas library

2.1. Data Cleaning

2.1.1. Handling Missing Values

The raw NASA data contained significant "noise" (NaN values) because not all detection methods yield both mass and radius data. We used `df.dropna()` to remove incomplete records.

2.1.2. Removing Duplicates

We observed that the raw dataset contained multiple entries for the same planet (e.g., *Kepler-452 b* appearing multiple times due to updated observations). To ensure our count of habitable worlds was accurate, we applied a filter (`drop_duplicates(subset=['pl_name'])`) to retain only one unique record per planet.

2.2. Data Subsetting (The "Goldilocks" Logic)

To tell the story of "Habitability," we created a subset of data based on standard astrobiological parameters:

- Distance: 0.8 to 1.5 AU (Astronomical Units) The distance where liquid water can exist.
- Size: 0.8 to 1.5 Earth Radii – The size range where a planet is likely rocky.
- Earth Similarity Sorting: To identify the best candidates for labeling, we calculated a "Distance Score" for each planet based on how close its coordinates were to Earth (1,1).

2.3. Exploratory Analysis

2.3.1. Attribute Selection

To determine the best variables for our visualization, we analyzed the completeness of every column in the dataset. This exploratory phase revealed that while Planetary Mass is scientifically relevant, it suffers from missing values compared to Planetary Radius. As shown in Figure 1, using Radius (`pl_rade`) allowed us to plot nearly 3,619 planets, whereas using Mass would have reduced our sample size. Based on this evidence, we selected Radius as the superior attribute for the Y-axis.

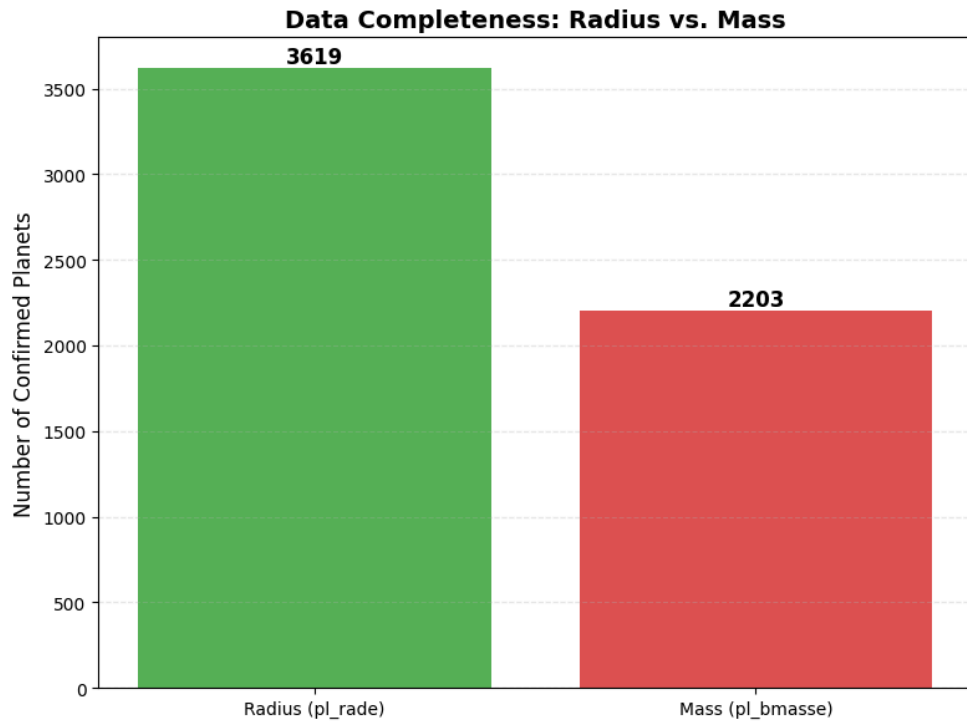


Figure 1: *Data Completeness Comparison. Radius data is far more complete than Mass data.*

2.3.2. Historical Context

We also plotted the cumulative discovery rate over time (Figure 2) to verify the "Volume" aspect of our dataset. The sharp exponential rise after 2010 (largely due to the Kepler Space Telescope) confirms that we are working with a modern, high-volume dataset suitable for "Big Data" analysis..

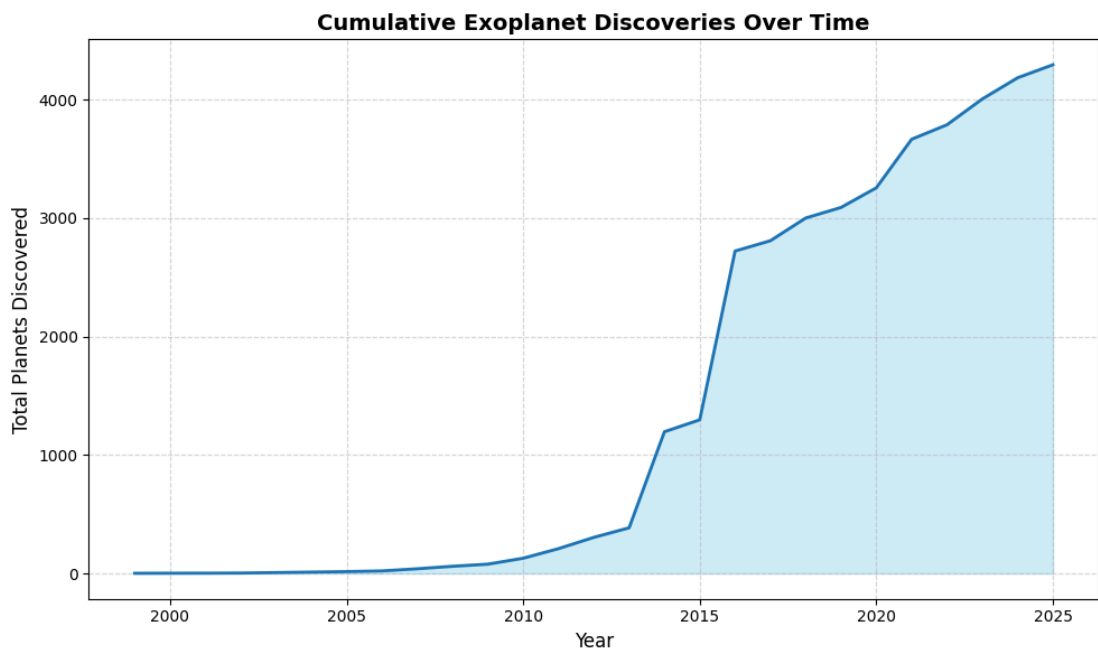


Figure 2: *Cumulative discoveries over time. The dataset volume has exploded in the last decade, providing a rich sample size for analysis.*

3. Visualization

3.1. Design Process

Our design goal was to visualize a massive disparity in scale. We needed to show the *abundance* of planets while simultaneously highlighting the *scarcity* of habitable ones.

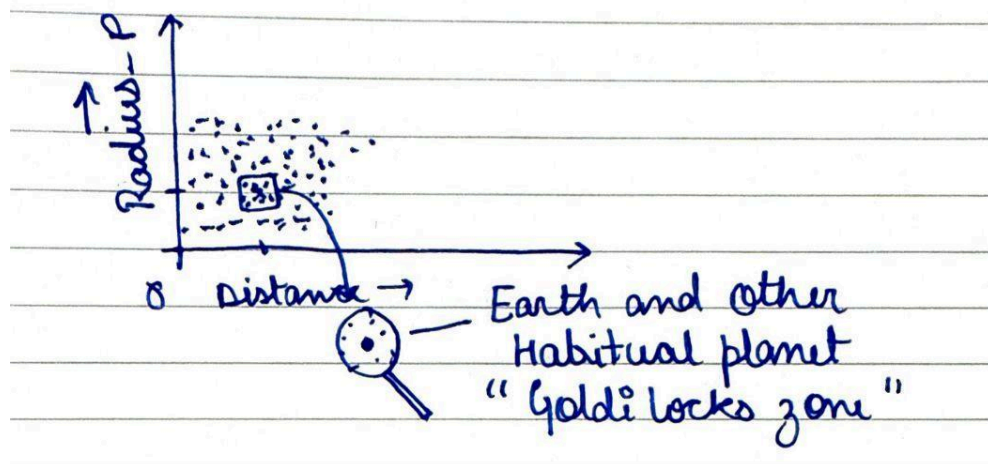


Figure 3: Initial sketch showing the concept of a main scatter plot with a "magnifying glass" effect to see the small Earth-like zone.

3.2. FINAL VISUALIZATION

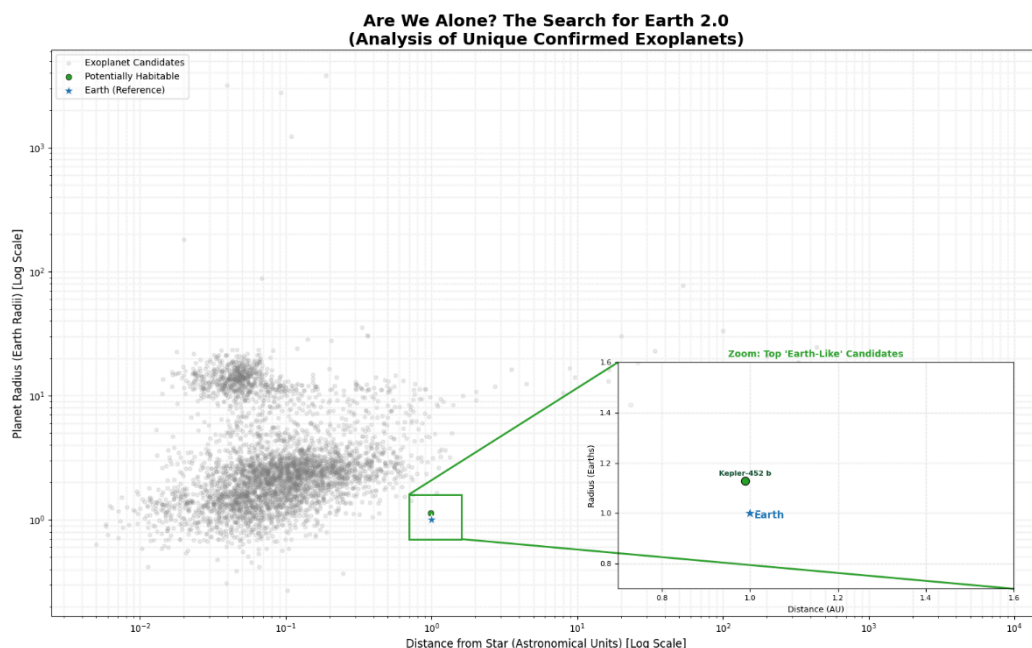


Figure 4: Log Scaled Scatter Plot of Unique Exoplanet Populations with Inset Zoom labeling

4. Justification of Design Choices

4.1. Chart Type

Scatter Plot (Log Scaled), we used a scatter plot because it was the best plot we could find to identify clusters in two-dimensional space.

- Problem: Planetary distances vary from 0.01 AU (scorched worlds) to 100 AU (frozen worlds). On a linear scale, the "Goldilocks Zone" would be compressed into a single pixel.

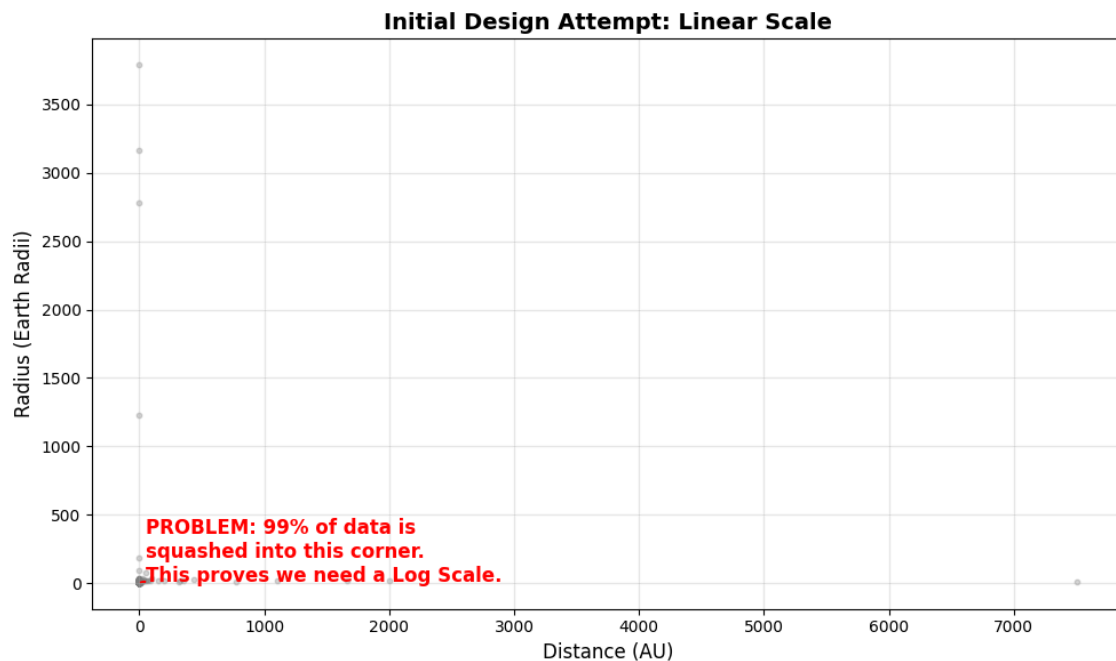


Figure 5: *Visualization without Log Scale 'Problem'*

- Solution: We applied a Logarithmic Scale. This expanded the "small numbers" region, allowing us to clearly distinguish between a planet at 0.5 AU and 1.0 AU

4.2. Measurement Units

We used Relative Units rather than absolute units (km) to make the graph intuitive:

- X-Axis (Astronomical Units): Defined as the distance from Earth to the Sun. A value of 1.0 places the viewer instantly in a familiar context.
- Y-Axis (Earth Radii): Defined relative to Earth's size. This allows the viewer to instantly judge if a planet is a "Super-Earth" (1.5x) or a "Gas Giant" (10x).

4.3. The Inset Plot

While the logarithmic scale effectively visualized the entire dataset, we implemented an additional inset plot using `mpl_toolkits` to enhance the viewer's experience. This 'Zoom-In' feature focuses specifically on the habitable region, allowing for easier reading of individual labels and details without losing the context.

- **Function:** This secondary plot focuses exclusively on the coordinates 0.7–1.6 (AU/Radii).
- **Labeling Strategy:** We programmatically labelled only unique planets closest to Earth's profile. We also labelled all habitable planets making the chart readable.

4.4. Colour

- **Context:** We coloured the general population Gray with high transparency ($\alpha=0.15$). This represents the volume out of the habitable zone.
- **Focus Data:** We coloured the habitable candidates Green (#2ca02c) with solid opacity. Green is semantically associated with life, making it the intuitive choice for habitable worlds.
- **Connection:** We used green connecting lines to link the inset box to the main plot. To represent the zoom in feature is related to habitable zone

5. Conclusion

Tools Used: We used Python programming language with libraries , Pandas handled the large-scale data cleaning, while Matplotlib was used for the advanced plotting features.

Critical Analysis: The visualization successfully answers the question "Are we alone?" by demonstrating that while Earth-like planets are statistically rare (a small green dot in the whole sea of gray), it definitely exists. The addition of labels for specific candidate *Kepler-452 b* added a tangible reality to the abstract data.

Collaboration: Kartik sourced the NASA dataset and researched the definitions of the Habitable Zone. Tanishq wrote the Python code, specifically implementing the all logic and the algorithm to see "Earth Similarity" amongst all planets. We collaborated on the final design layout and report writing.

6. References

[1] **Dataset:** NASA Exoplanet Archive (2025). *Planetary Systems Composite Data*. California Institute of Technology. DOI: [10.26133/NEA12](https://doi.org/10.26133/NEA12)

Christiansen et al. (2025). *The NASA Exoplanet Archive and Exoplanet Follow-up Observing Program: Data, tools, and usage*. **The Planetary Science Journal**, 6, 186. <https://doi.org/10.3847/PSJ/ade3c2>

[2] [CSC1143/CSC1175 Data Visualisation Assignment 202509](#)

Tools: *Matplotlib: Visualization with Python*, Pandas